MATERIALS 2

Lecture 3: Phase Diagrams II

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Definitions

- An **eutectic system** is an alloy of two or more metals that melts or solidifies at a single temperature that is lower than the melting point of any of the constituents.

- The **eutectic point** (or temperature) is the lowest possible melting temperature for which all of the mixing ratios of the component substances are in an eutectoid.

- The **eutectic solidification** is defined as:

$$\text{Liquid (L)} \xrightarrow[\text{cooling}]{} \alpha\text{-solid solution} + \beta\text{-solid solution}$$

- **Hypoeutectic** and **Hypereutectic systems** are systems that could form an eutectoid, but do not have the appropriate ratio of component substances.

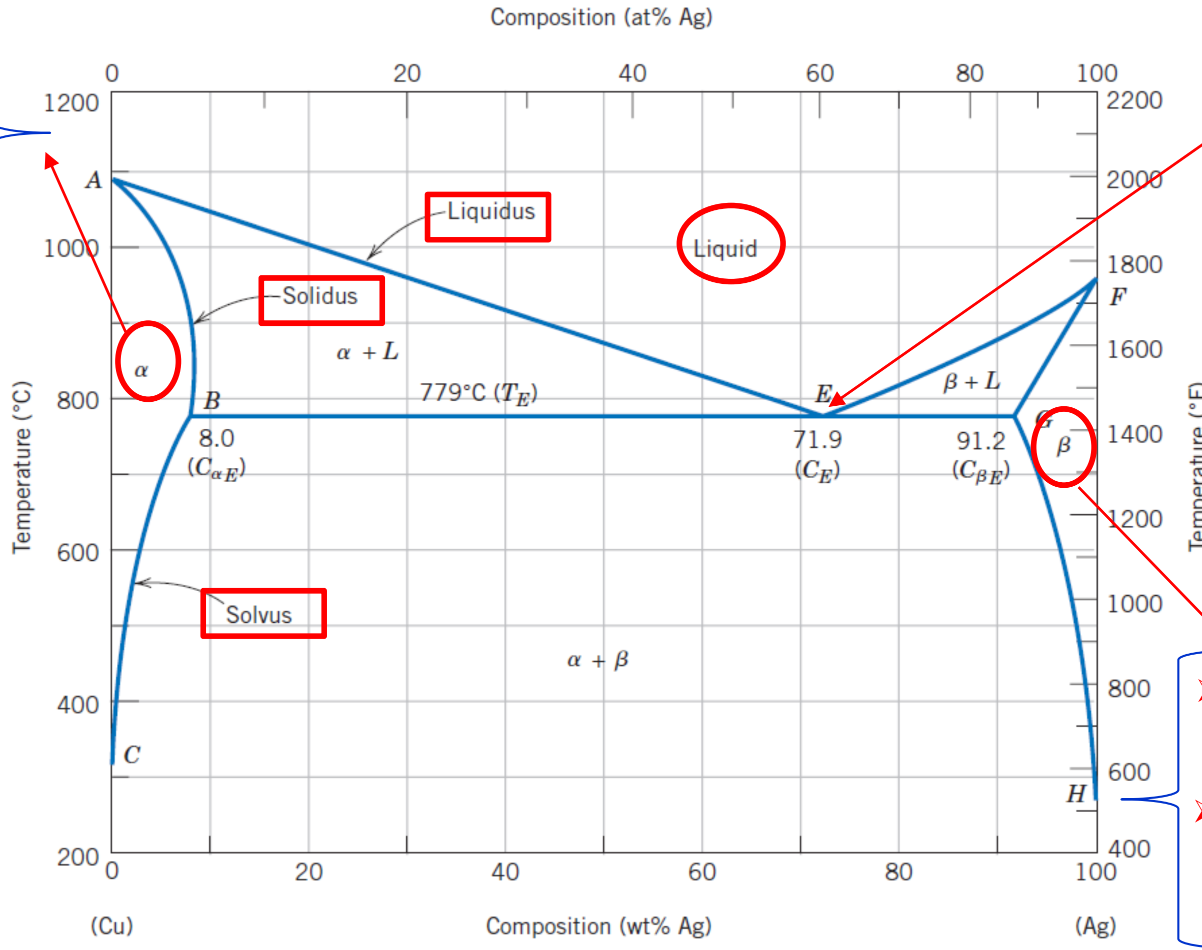
- A **hypoeutectic system** has a lower percentage of β and a higher percentage of α than an eutectic composition.

- A **hypereutectic system** has a higher percentage of β and a lower percentage of α than a eutectic composition.

Binary Eutectic Systems

Copper-Silver Phase Diagram

- It is a solid-solution rich in copper.
- It has silver as the solute component and an FCC crystal structure.



Eutectic point

- It is a solid-solution rich in silver.
- It has an FCC crystal structure and copper as the solute component.

Eutectic reaction at T_E: $L(C_E) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha(C_{\alpha E}) + \beta(C_{\beta E})$, $L(71.9wt\%Ag) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha(8wt\%Ag) + \beta(91.2wt\%Ag)$

Calculating Phase Fraction

Example: Lead-Tin Phase Diagram

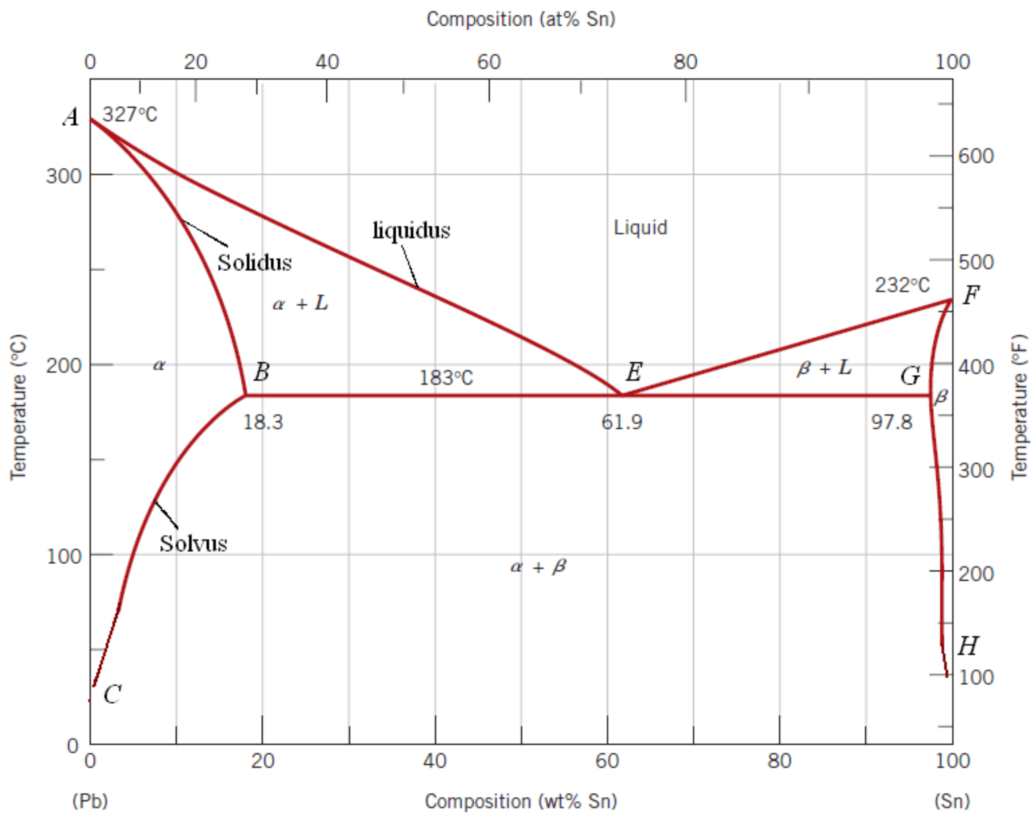
For a 40 wt% Sn–60 wt% Pb alloy at 150°C

a) What phase(s) is (are) present?

b) What is (are) the composition(s) of the phase(s)?

c) Calculate the relative mass fraction of each phase. What do you conclude?

d) Calculate the volume fraction of each phase. At 150°C, take the densities of Pb and Sn to be 11.23 and 7.24 g/cm3, respectively. What do you conclude?

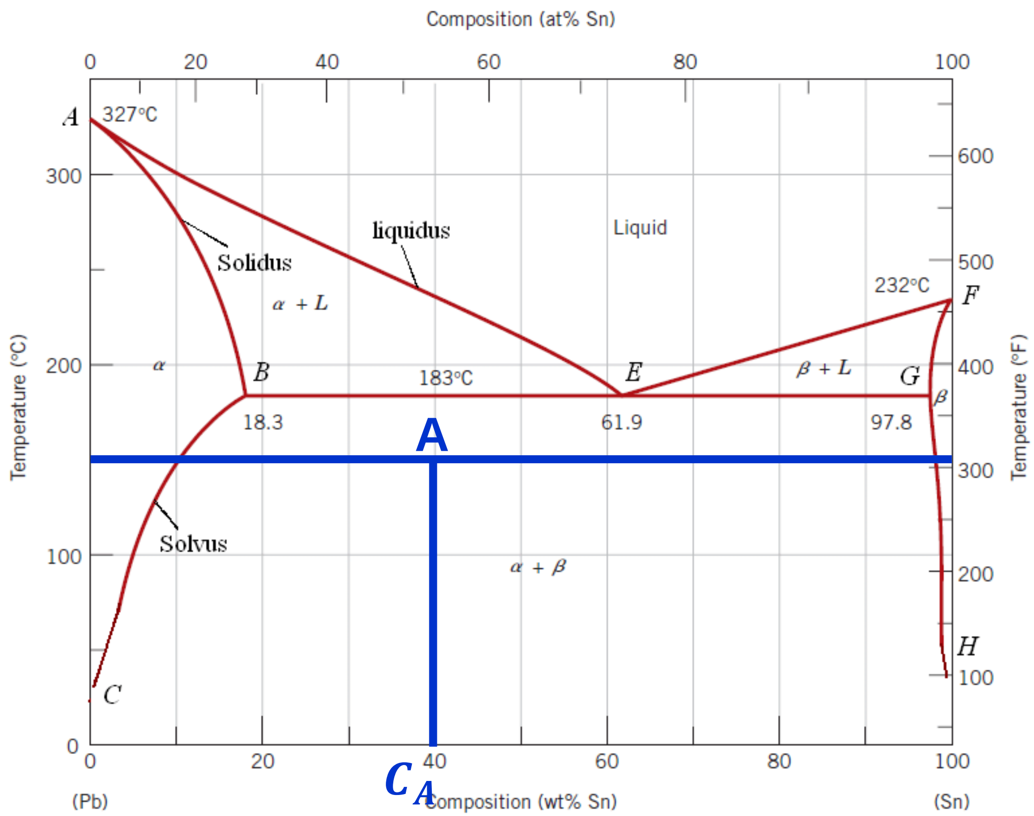


Calculating Phase Fraction

Example: Lead-Tin Phase Diagram

a) What phase(s) is (are) present?

- Draw a **tie line** at 150°C
- Locate the temperature–composition point on the phase diagram (point **A**).
- Point **A** is within the $\alpha + \beta$ region, so both α and β phases will coexist. .



Calculating Phase Fraction

Example: Lead-Tin Phase Diagram

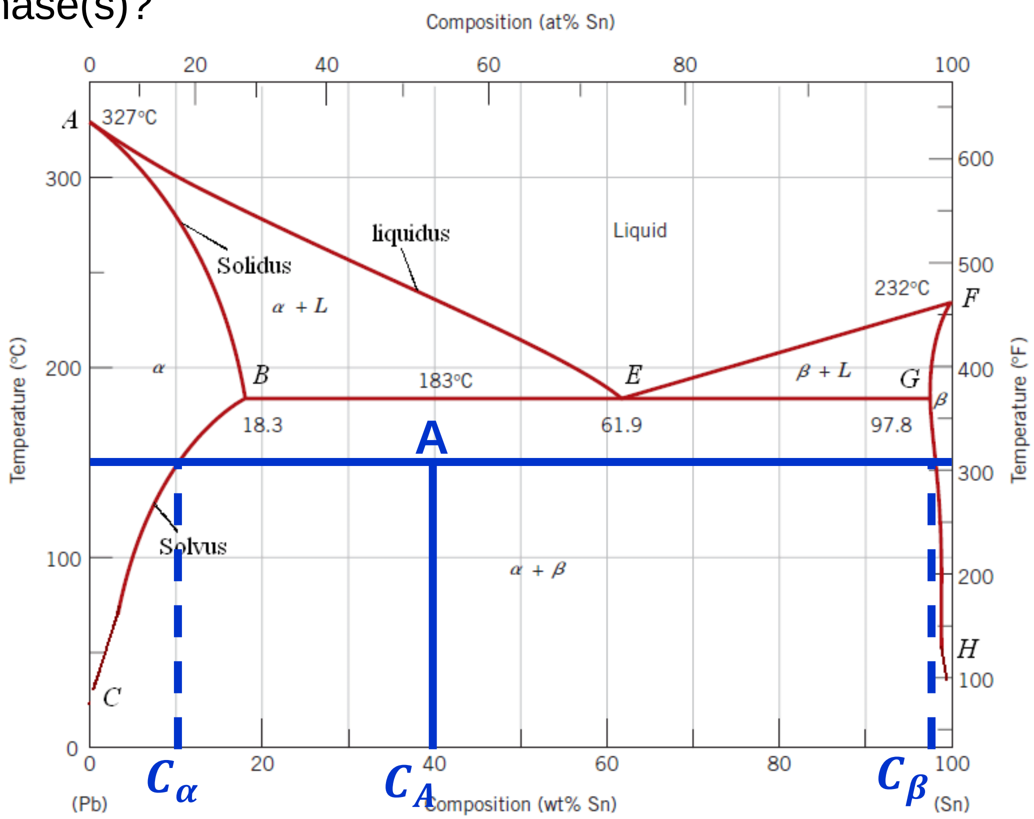
b) What is (are) the composition(s) of the phase(s)?

- The composition of the α phase corresponds to the intersection between the tie line and the $\alpha/(\alpha+\beta)$ solvus phase boundary:

$$C_\alpha = 11\ wt\%\ Sn\text{–}\ 89\ wt\%\ Pb$$

- The composition of the β phase corresponds to the intersection between the tie line and the$\beta/(\alpha+\beta)$ solvus phase boundary:

$$C_\beta = 98\ wt\%\ Sn\text{–}\ 2\ wt\%\ Pb$$



Calculating Phase Fraction

Example: Lead-Tin Phase Diagram

c) Calculate the relative mass fraction of each phase. What can you conclude?

- Using the lever rule:

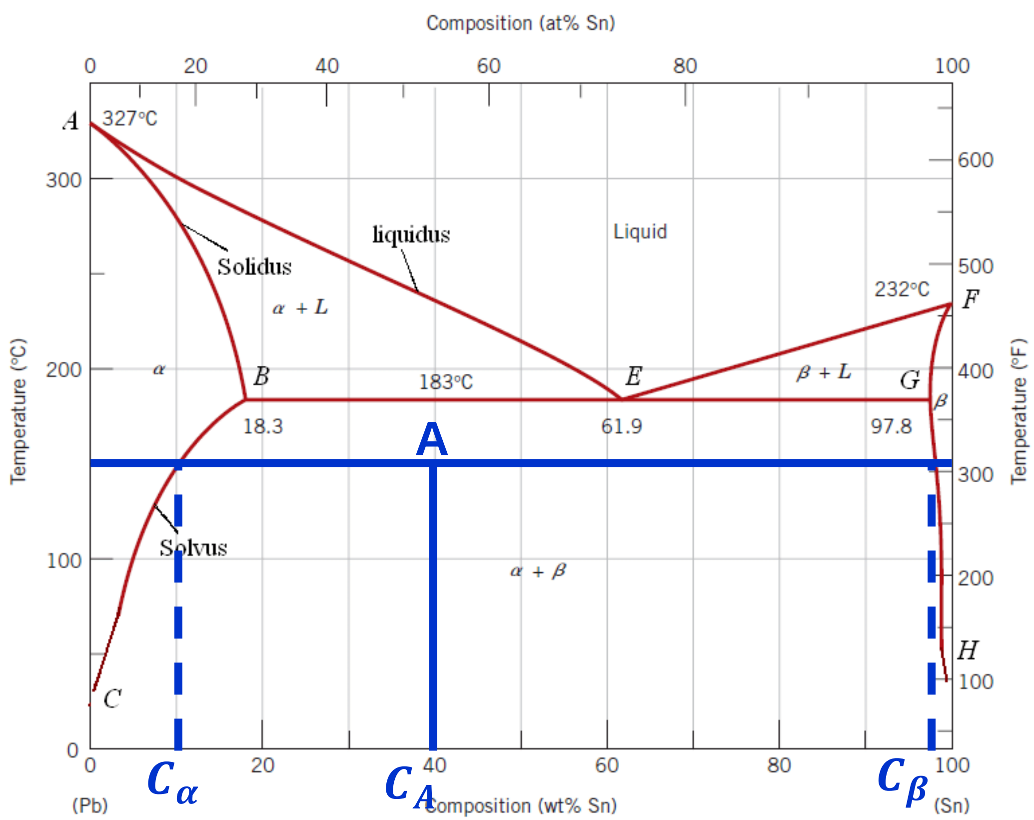
$$W_\alpha = \frac{C_\beta - C_A}{C_\beta - C_\alpha}$$

$$\Rightarrow W_\alpha = \frac{98wt\%Sn - 40wt\%Sn}{98wt\%Sn - 11wt\%Sn} = 0.67$$

$$W_\beta = \frac{C_A - C_\alpha}{C_\beta - C_\alpha}$$

$$\Rightarrow W_\beta = \frac{40wt\%Sn - 11wt\%Sn}{98wt\%Sn - 11wt\%Sn} = 0.33$$

- Conclusion: $W_\alpha + W_\beta = 1$



Calculating Phase Fraction

Example: Lead-Tin Phase Diagram

d) Calculate the volume fraction of each phase. At 150°C, take the densities of Pb and Sn to be 11.23 and 7.24 g/cm^3, respectively.

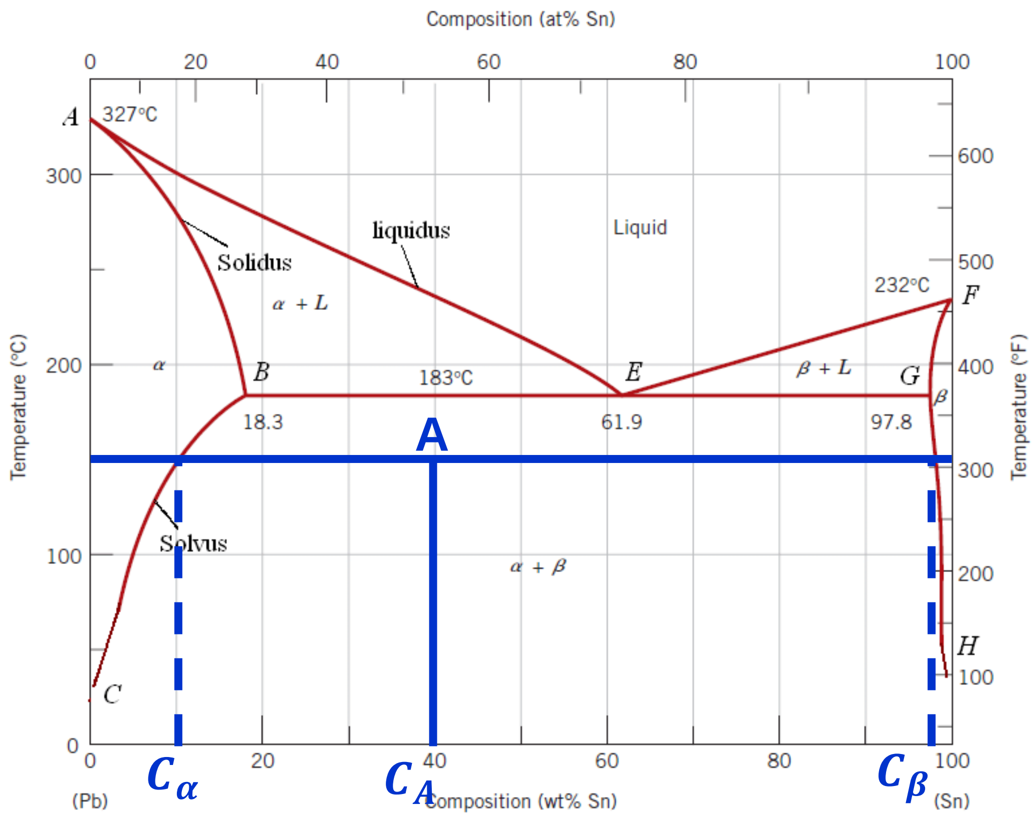
- In order to calculate the volume fractions of each phase, it is necessary to determine the density of each phase

- $$\rho_\alpha = \frac{100}{\frac{C_{\alpha(\text{Sn})}}{\rho_{\text{Sn}}}+\frac{C_{\alpha(\text{Pb})}}{\rho_{\text{Pb}}}}$$

$$\Rightarrow \rho_\alpha = \frac{100}{\frac{11}{7.24\ g/cm^3}+\frac{89}{11.23\ g/cm^3}} = 10.59 g/cm^3$$

- $$\rho_\beta = \frac{100}{\frac{C_{\beta(\text{Sn})}}{\rho_{\text{Sn}}}+\frac{C_{\beta(\text{Pb})}}{\rho_{\text{Pb}}}}$$

$$\Rightarrow \rho_\beta = \frac{100}{\frac{98}{7.24\ g/cm^3}+\frac{2}{11.23\ g/cm^3}} = 7.29 g/cm^3$$



Calculating Phase Fraction

Example: Lead-Tin Phase Diagram

d) Calculate the volume fraction of each phase. At 150°C, take the densities of Pb and Sn to be 11.23 and 7.24 g/cm^3, respectively. What do you conclude?

- Using the conversion of mass fractions of α and β phases to volume fractions:

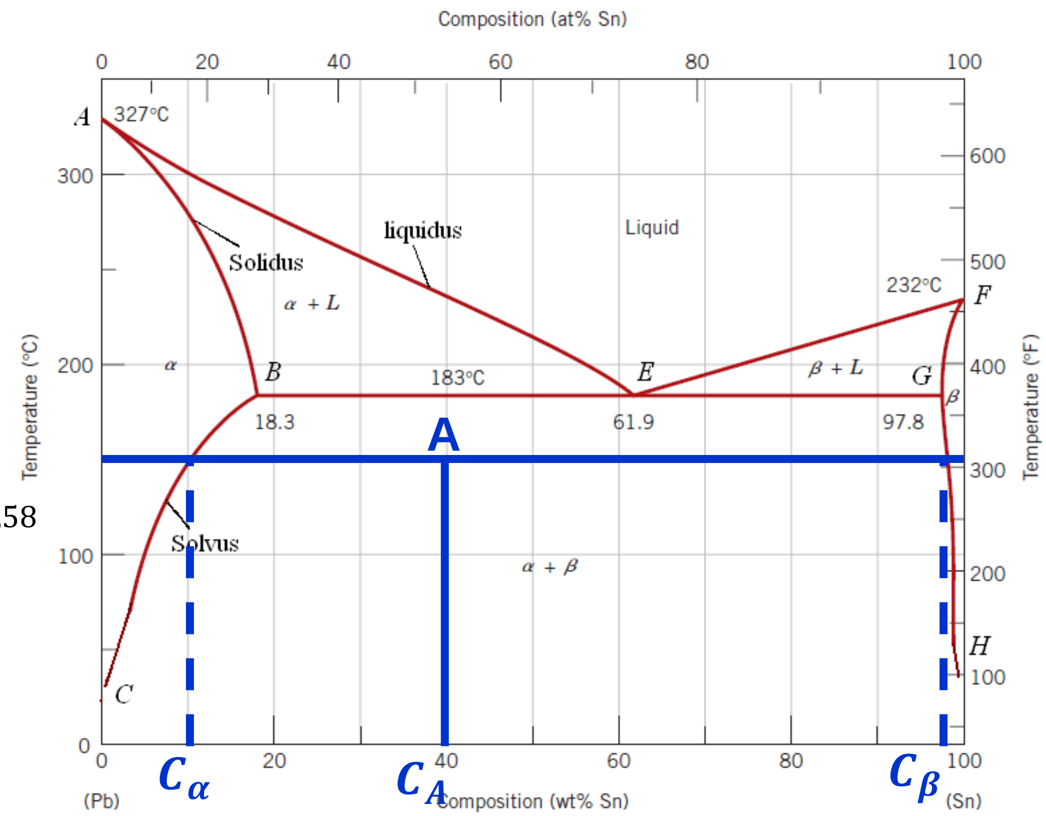
- The volume fraction of the α phase, V_α:

$$V_\alpha = \frac{\frac{W_\alpha}{\rho_\alpha}}{\frac{W_\alpha}{\rho_\alpha}+\frac{W_\beta}{\rho_\beta}} \Rightarrow V_\alpha = \frac{\frac{0.67}{10.59g/cm^3}}{\frac{0.67}{10.59g/cm^3}+\frac{0.33}{7.29g/cm^3}} = 0.58$$

- The volume fraction of the β phase, V_β:

$$V_\beta = \frac{\frac{W_\beta}{\rho_\beta}}{\frac{W_\alpha}{\rho_\alpha}+\frac{W_\beta}{\rho_\beta}} \Rightarrow V_\beta = \frac{\frac{0.33}{7.29g/cm^3}}{\frac{0.67}{10.59g/cm^3}+\frac{0.33}{7.29g/cm^3}} = 0.42$$

- Conclusion: $V_\alpha + V_\beta = 1$



Development of Microstructure in Eutectic Systems

- A **microstructure** is a very small scale structure of a material, defined as the structure of a prepared surface of material as revealed by an optical microscope above 25× magnification.

- A **microstructural observation** is generally performed using optical or scanning electron microscopes to magnify features of the material under analysis.

- The **microstructure of a material** (metals, polymers, ceramics, or composites) can strongly influence the physical properties of the material, such as strength, toughness, ductility, hardness, corrosion resistance, high/low temperature behavior or wear resistance.

- For many materials, different **phases** can exist at the same time.

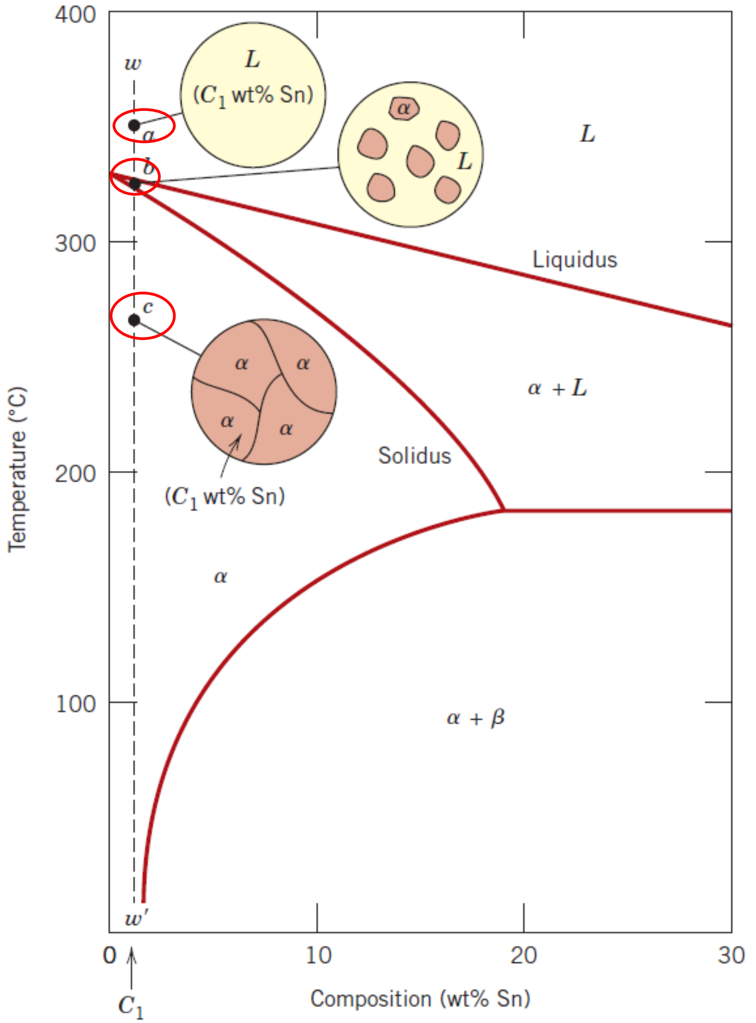
- Depending on the composition, several types of **microstructures** are possible for the **slow cooling** of alloys belonging to binary eutectic systems.

Slow Cooling-Equilibrium

<u>Case 1</u>: Compositions ranging between a pure component and the maximum solid solubility for that component at room temperature (20°C)

<u>Example</u>: Lead-rich alloys containing between 0 and about 2 wt% Sn (for the α-phase solid solution) and also between 99 wt% Sn and pure tin (for the β-phase).

- Consider an alloy of composition C_1 as it is **slowly cooled** (line ww') from a temperature of **350°C** within the **liquid-phase region**. (point a)
- The alloy remains totally liquid and of composition C_1 until the **liquidus line** is crossed at **330°C**, where the solid α-phase begins to form.
- While passing through the narrow $\alpha + L$ phase region, **solidification** begins, with continued cooling, where more of the solid α is formed. (point b)
- The **liquid- and solid-phase compositions** are **different**, which follow along the liquidus and solidus phase boundaries, respectively.
- **Solidification** reaches completion at the point where ww' crosses the solidus line.
- The resulting alloy is **polycrystalline** with a **uniform composition** of C_1, and there will not be any subsequent changes occur upon cooling to room temperature. (point c)

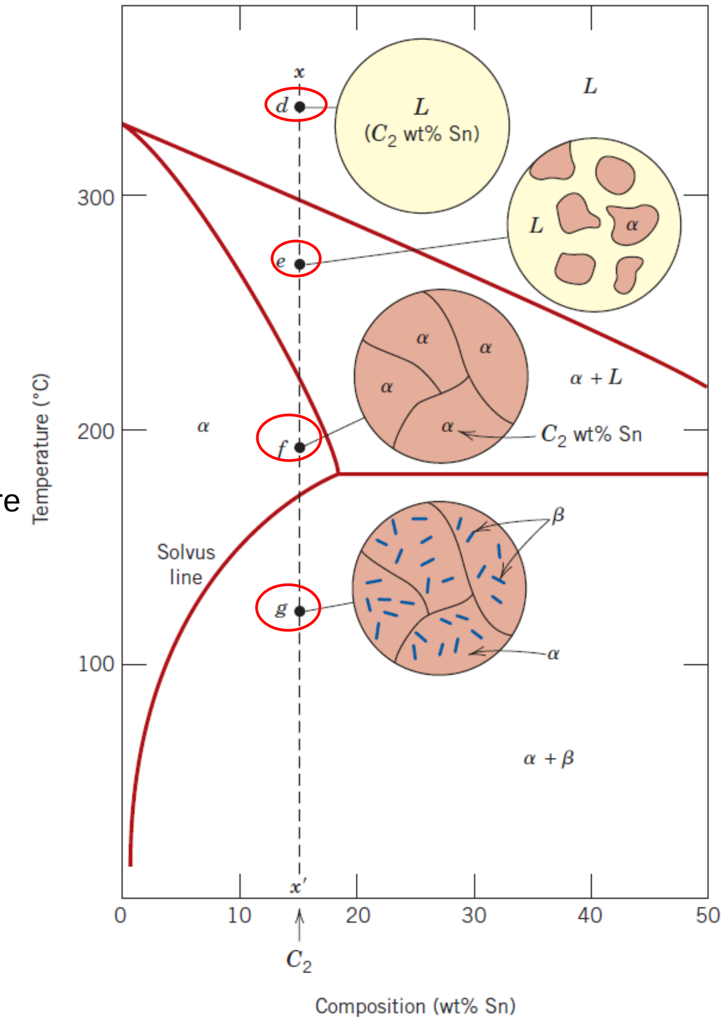


Slow Cooling-Equilibrium

Case 2: Compositions that range between the room temperature solubility limit and the maximum solid solubility at the eutectic temperature

Example: Lead–tin system with compositions extend from about 2 to 18.3 wt% Sn (for lead-rich alloys) and from 97.8 to approximately 99 wt% Sn (for tin-rich alloys).

- Consider an alloy of composition C_2 as it is **cooled** along the vertical line xx'.
- While going down to the intersection of xx' and the **solvus line**, changes that occur are similar to the previous case as we pass through the corresponding phase regions (points d, e, and f).
- Just above the **solvus line** intersection with the xx' (point f), the microstructure consists of a grains of composition **C2**.
- Upon crossing the **solvus line**, the **α-solid solubility** is exceeded, which results in the formation of small **β-phase particles**; these are indicated in the microstructure inset at point g.
- With **continued cooling**, these particles grow in size because the mass fraction of the **β- phase** increases slightly with decreasing temperature.



Slow Cooling-Equilibrium

<u>Case 3</u>: Solidification of the eutectic composition

<u>Example</u>: Lead–tin alloy with 61.9 wt% Sn having a composition C_3

- Consider alloy having **C3** as a composition that is cooled from a temperature within the liquid-phase region (**250°C**) down the vertical line yy' (point h).

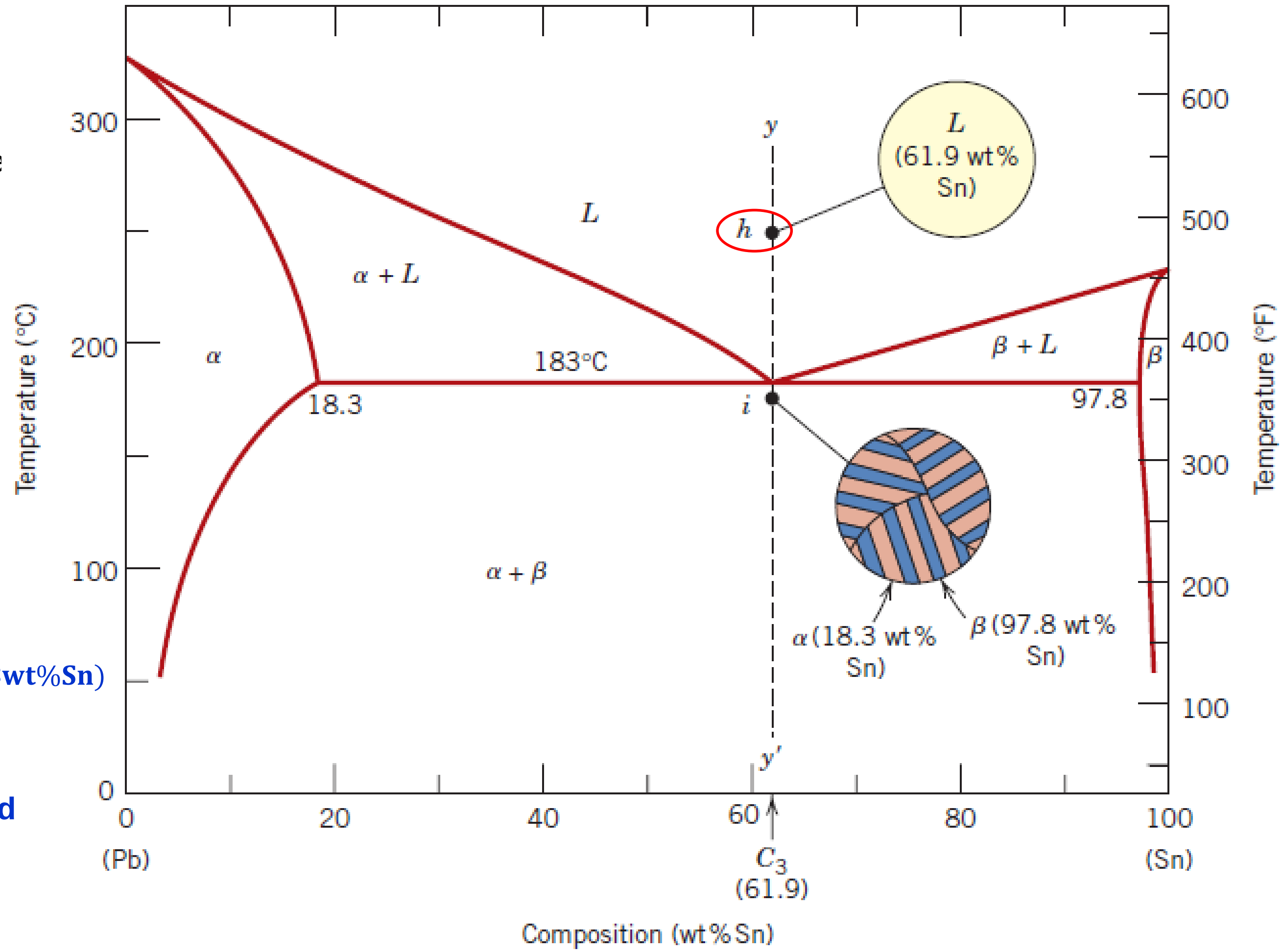
- As the temperature is lowered, there is no changes until we reach the **eutectic temperature**, **183°C**.

- Upon crossing the **eutectic point**, the **liquid** transforms into the two α and β phases.

- **Eutectic reaction** :

$$L(61.9wt\%Sn) \underset{heating}{\overset{cooling}{\rightleftharpoons}} \alpha(18.3wt\%Sn) + \beta(97.8wt\%Sn)$$

- During this transformation, there must be a **redistribution of the lead and tin components** because the two α and β phases have different compositions.

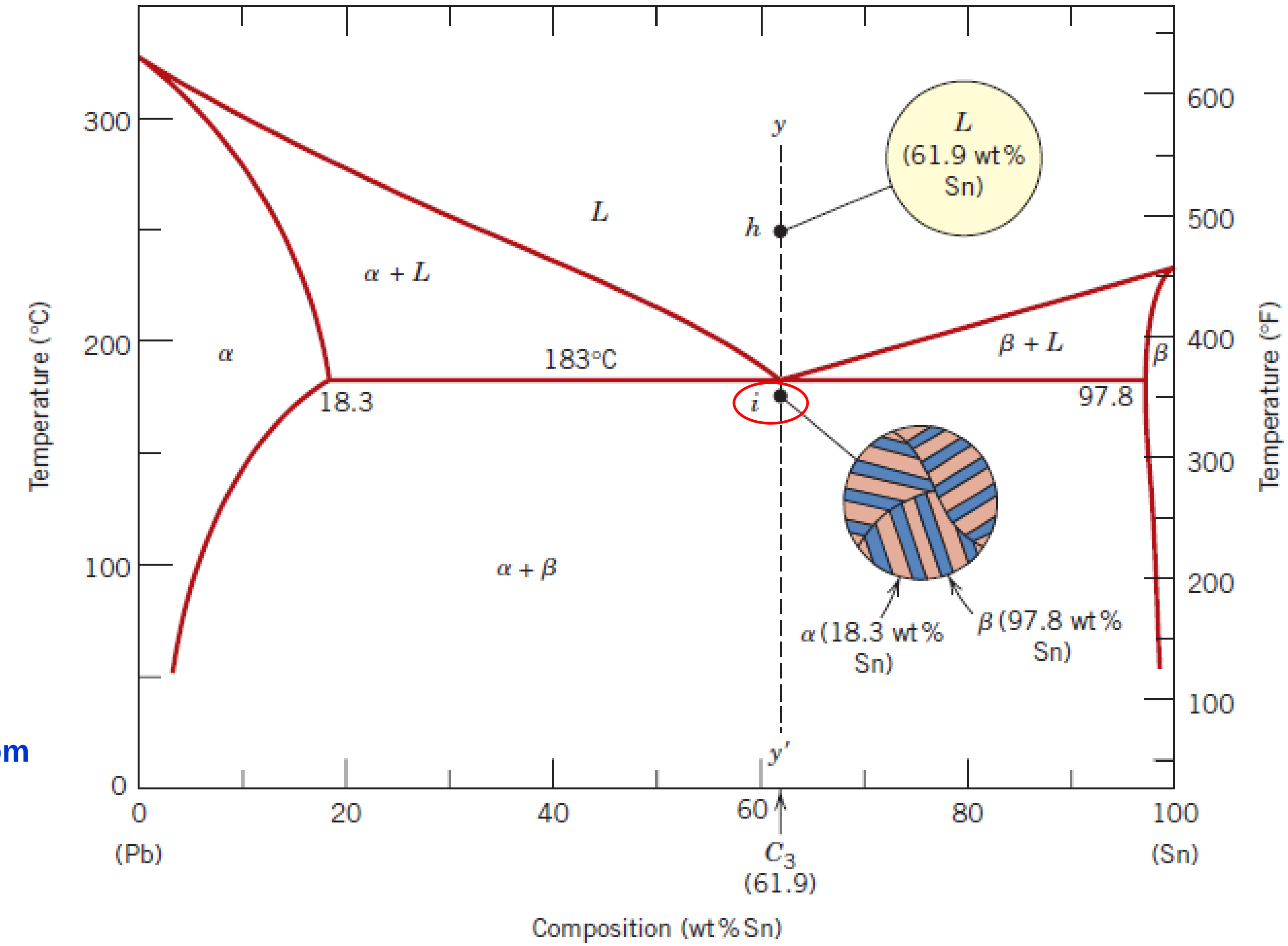


Slow Cooling-Equilibrium

Case 3: Solidification of the eutectic composition

Example: Lead–tin alloy with 61.9 wt% Sn having a composition C_3

- This redistribution is accomplished by **atomic diffusion**.
- During the transformation, the microstructure of the solid that results from this transformation consists of **alternating layers** (or **lamellae**) of the α and β phases.
- This microstructure (point i) is called a **eutectic structure**.
- **Subsequent cooling** of the alloy from just below the **eutectic to room temperature** results in only **minor microstructural alterations**.

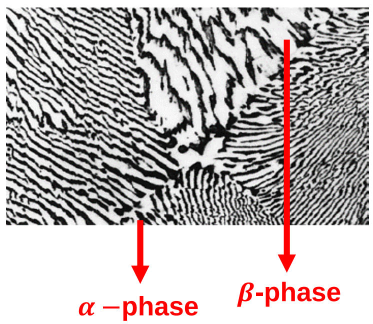


Slow Cooling-Equilibrium

<u>Case 3</u>: Solidification of the eutectic composition

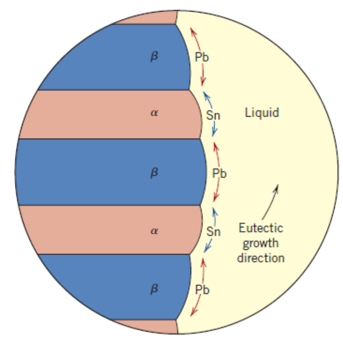
<u>Example</u>: Lead–tin alloy with 61.9 wt% Sn having a composition $\boldsymbol{C_3}$

- Photomicrograph of the **Eutectic Structure**:



This microstructure consists of alternating layers of a **lead-rich α -phase solid solution** (dark layers), and a **tin-rich β -phase solid solution** (light layers).

- Microstructural change during **the eutectic transformation**:



α-β layered eutectic growing into and replacing the **liquid phase**

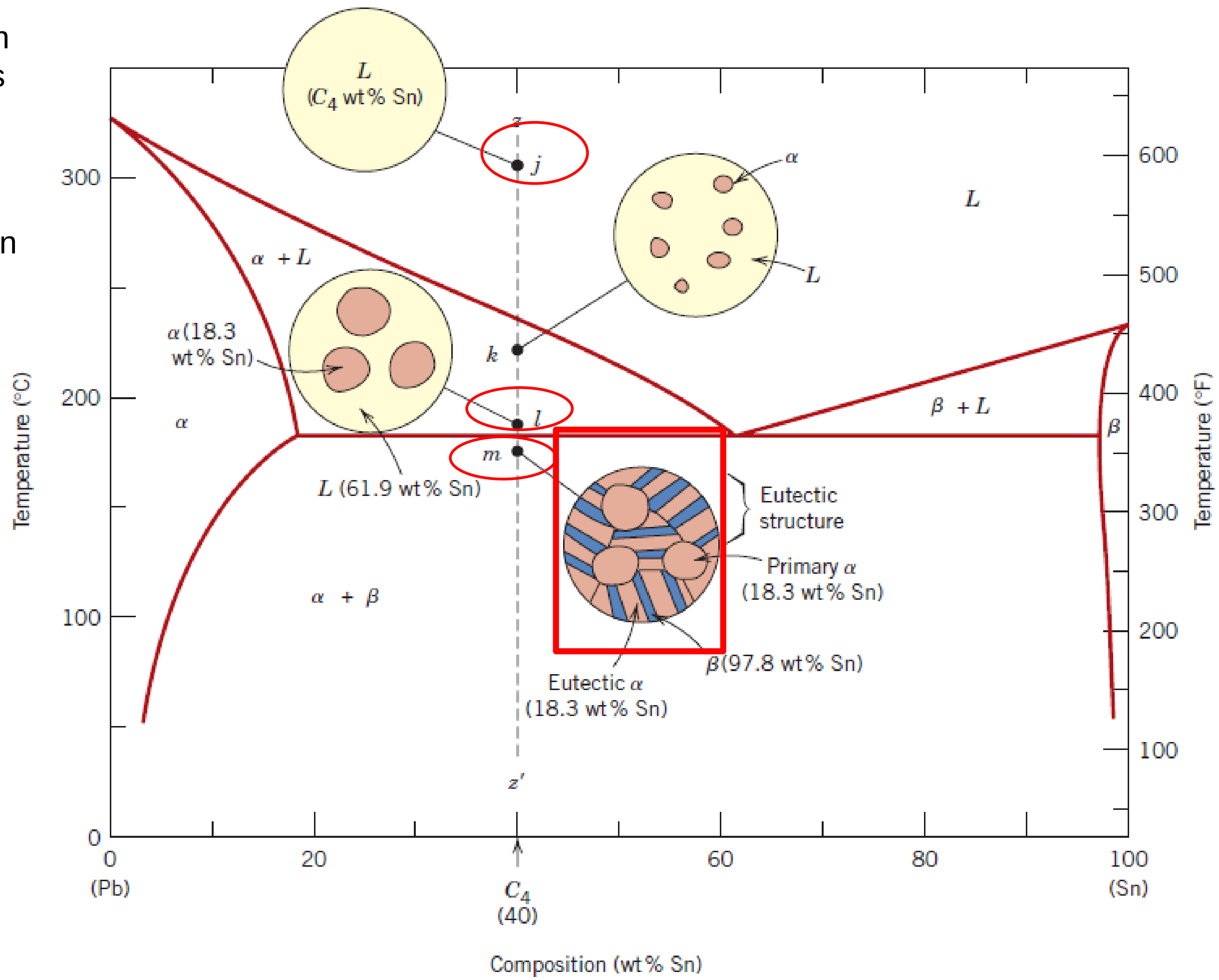
- The process of the **redistribution of lead and tin** occurs by **diffusion in the liquid** just after the eutectic–liquid interface.

- The arrows indicate the directions of diffusion of lead and tin atoms; **lead atoms** diffuse toward the **α-phase layers** because this **α phase is lead-rich** (18.3 wt% Sn–81.7 wt% Pb); conversely, the direction of diffusion of **tin atoms** is in the direction of the **β phase**, which is **tin-rich** (97.8 wt% Sn–2.2 wt% Pb) layers.

Slow Cooling-Equilibrium

Case 4: All compositions other than the eutectic that, when cooled, cross the eutectic isotherm.

- Consider lead-tin alloy with a composition C_4 which lies to the left of the eutectic; as the temperature is lowered, we move down the line zz', beginning at point j.
- The microstructural development between points j and l is similar to that for case 2.
- Just **prior to the eutectic isotherm** (point l), the **α and liquid phases** are present with compositions of **18.3 and 61.9 wt% Sn**, respectively.
- As the temperature is lowered to just **below the eutectic isotherm** (point m), the **liquid phase**, which is of the eutectic composition, transforms into the **eutectic structure** (which is also called alternating **α and β lamellae**).
- The microstructure at point m contains **α-phase** in both the **eutectic structure** and the phase that is formed while cooling through the **(α + L)-phase field**.

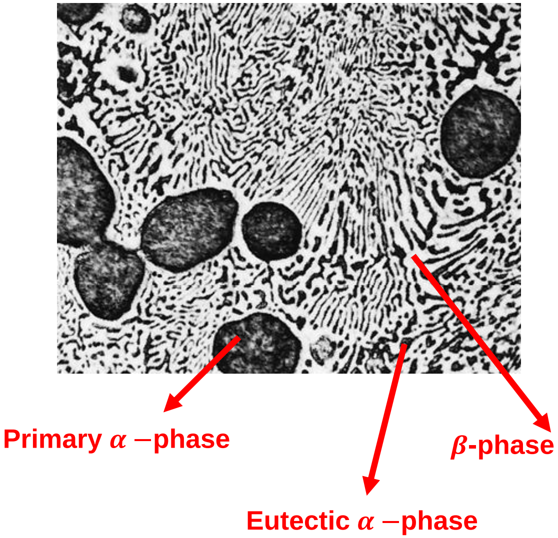


❖ The one that resides in the **eutectic structure** is called **eutectic α**, whereas the other that formed prior **due to crossing the eutectic isotherm** is termed **primary α**.

Slow Cooling-Equilibrium

Case 4: All compositions other than the eutectic that, when cooled, cross the eutectic isotherm.

- Photomicrograph of the microstructure (point m) of a lead–tin alloy of composition 50 wt% Sn–50 wt% Pb :



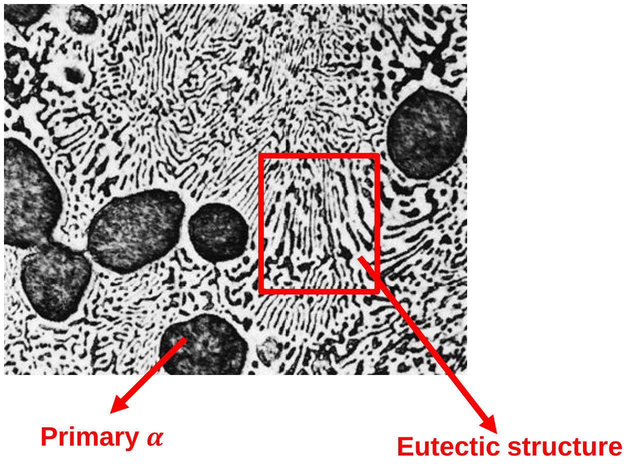
- This microstructure is composed of a **primary lead-rich α-phase** (>large dark regions) within a **lamellar eutectic structure** consisting of a **tin-rich β phase** (light layers) and **a lead-rich α phase** (dark layers).

Microconstituents

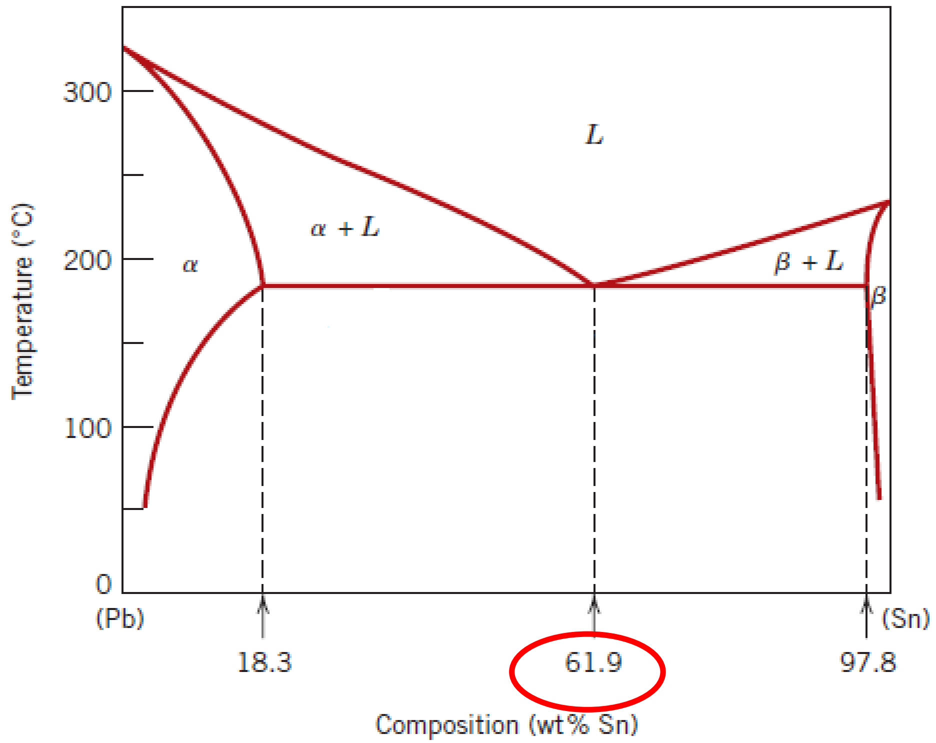
- A **microconstituent** is an element of the microstructure having an identifiable and characteristic structure.

- For example, at point m, there are two microconstituents:
  1. **primary α**
  2. **eutectic structure**

- Thus, the **eutectic structure** is a **microconstituent** even though it is a mixture of two phases due to its distinct **lamellar structure** with a fixed ratio of the two phases.



Relative Amounts



- The **eutectic microconstituent** may be assumed to have a composition of 61.9 wt% Sn since this microconstituent is always formed from the **liquid** having the **eutectic composition**.

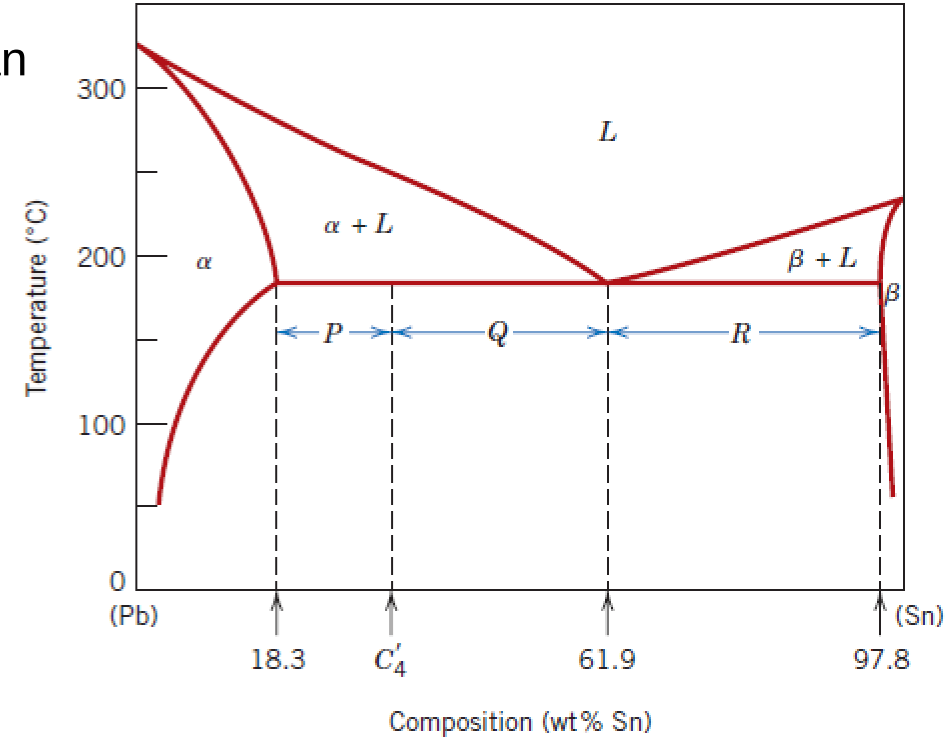
Relative Amounts

- Example: consider the Pb-Sn alloy of composition C'_4

- Using the **lever rule**: The **mass fraction** of the **eutectic microconstituent** W_e can be determined from the **tie line** between the α**-(**$\alpha + \beta$**)** phase boundary (18.3 wt% Sn) and the **eutectic composition** (61.9 wt% Sn):

$$W_e = W_L = \frac{P}{P + Q}$$

$$W_e = W_L = \frac{C'_4 - 18.3}{61.9 - 18.3}$$

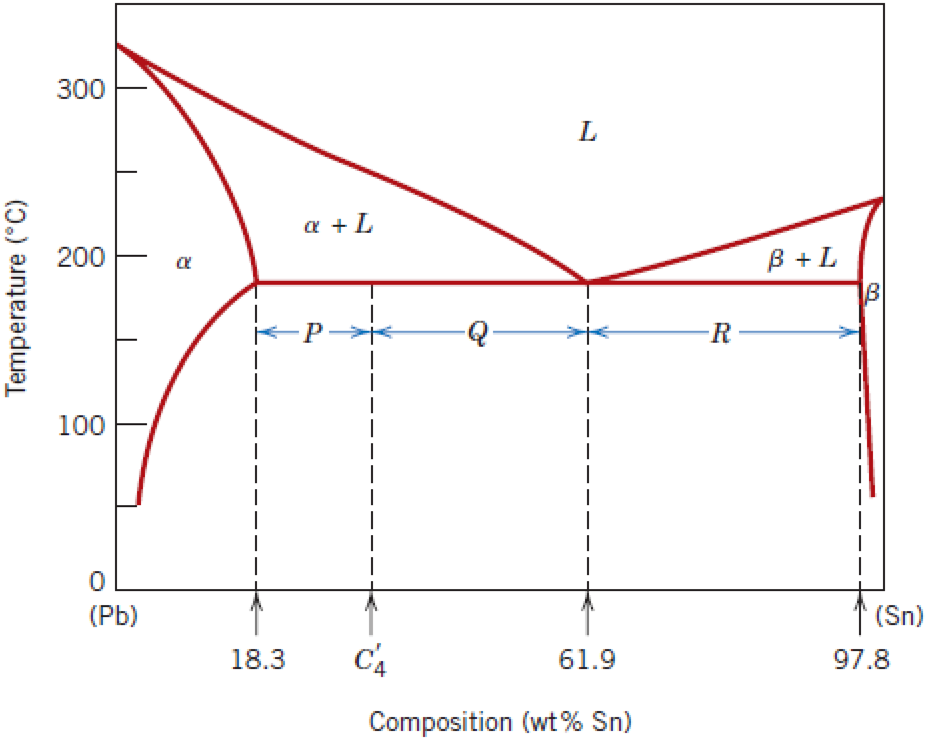


Relative Amounts

- The **mass fraction** of **primary α**, $W_{\alpha'}$, is the mass fraction of the α phase that existed prior to the eutectic transformation can be found as :

$$W_{\alpha'} = \frac{Q}{P+Q}$$

$$W_{\alpha'} = \frac{61.9 - C'_4}{61.9 - 18.3}$$

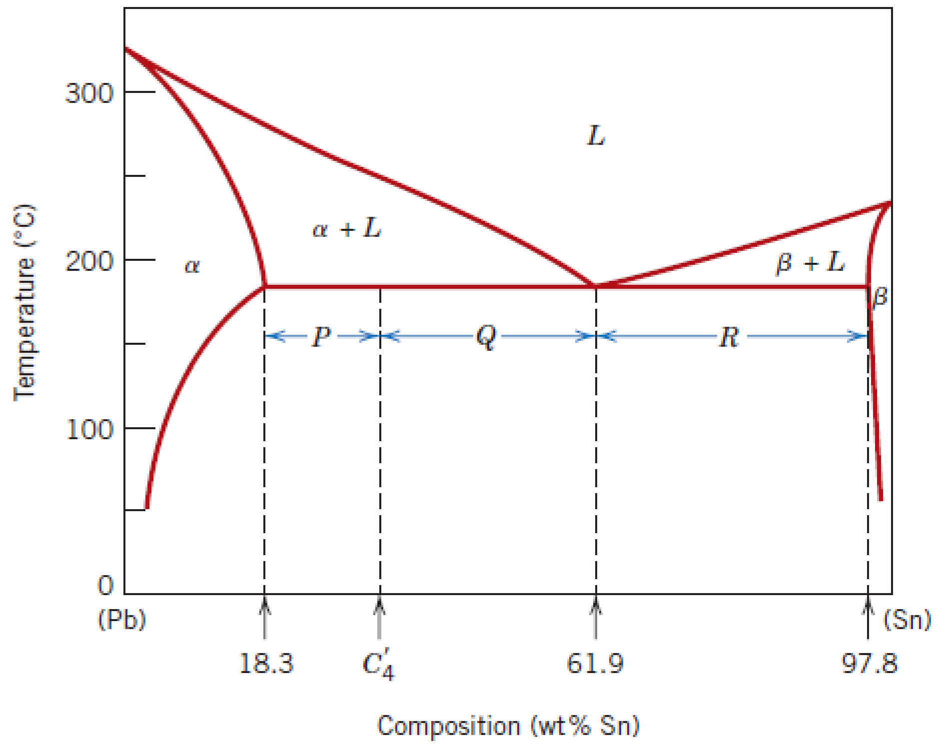


Relative Amounts

- The mass fractions of **total α**, W_α (both **eutectic and primary α**) is determined by use of the **lever rule** and a tie line that extends entirely across the α **+β** phase field as:

$$W_\alpha = \frac{Q+R}{P+Q+R}$$

$$W_\alpha = \frac{97.8 - C'_4}{97.8 - 18.3}$$

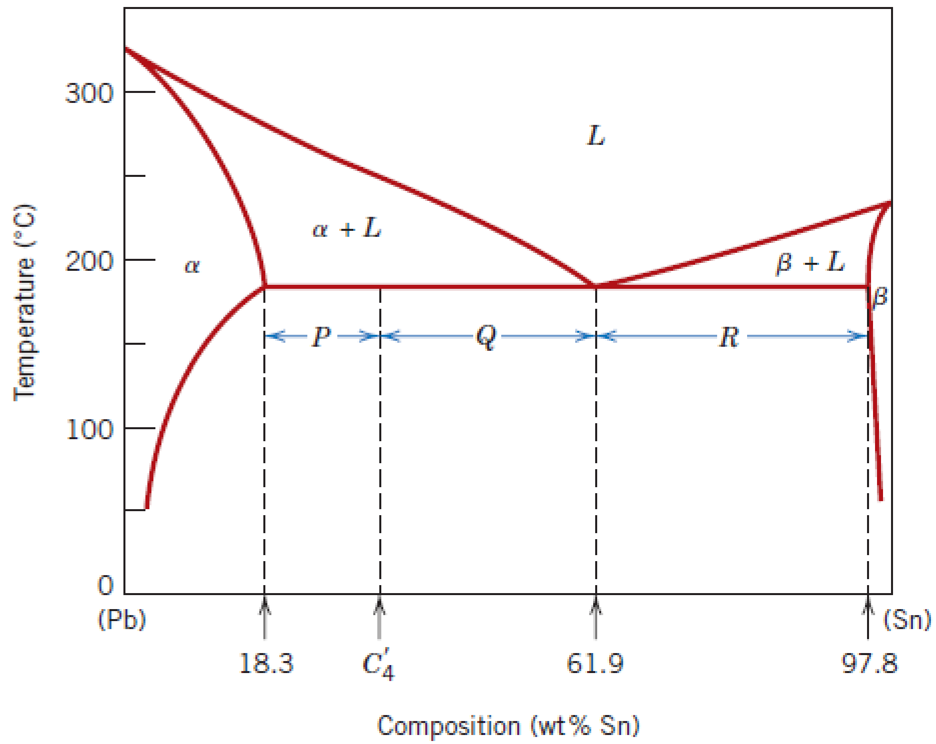


Relative Amounts

- The mass fractions of **total $\boldsymbol{\beta}$** , $\boldsymbol{W_\beta}$, is found using the **lever rule** and a tie line that extends entirely across the $\boldsymbol{\alpha+\beta}$ phase field as:

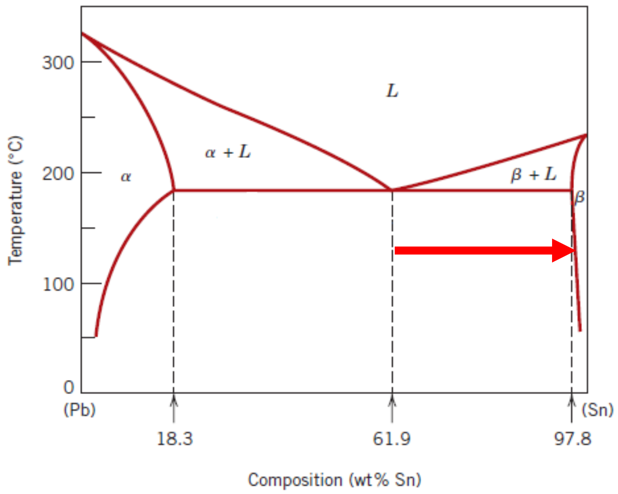
$$W_\beta = \frac{P}{P+Q+R}$$

$$W_\beta = \frac{C'_4 - 18.3}{97.8 - 18.3}$$



Microstructures

- Analogous transformations and microstructures result for alloys having compositions to the **right of the eutectic** (i.e., between 61.9 and 97.8 wt% Sn). However, **below the eutectic temperature**, the microstructure will consist of the **eutectic** and **primary β** microconstituents because, upon cooling from the liquid, we pass through the **β +L** phase field.



- For case 4, if the **conditions of equilibrium** are not maintained while passing through the α (or β) **+L** phase region, some consequences will result on the microstructure upon crossing the eutectic isotherm:

  1. The grains of the primary microconstituent will be cored, that is, have a non-uniform distribution of solute across the grains.
  2. The mass fraction of the eutectic microconstituent formed will be greater than that at the equilibrium situation.

Thank you for your attention!